Traffic engg in computer networks

module-1 1m

1. what is a network protocol
2. what is Data rate or bandwitdh
3. what is jitter
4. give atleast 2 reasons for packet loss in general
5. what are ping and traceroute tools
6. what is transmission and propagation delays
7. what terms do you need to calculate propagation delay over a wire
8. what terms do you need to calculate transmission delay over a wire
9. explain the terms : attenuation, noise, interference
10. how to use iperf3 in linux to calculate the network performance metrics between 2 machines

module-1 2.5m

1. list all the protocols in OSI layer
2. what is digital data transmisison preferred over analog transmission
3. difference between amplifier and repeater. What does repeater better than amplifier for digital transmission
4. explain switched ethernet, and how does it build the MAC address tables within the LAN
5. what is the problem with sending a broadcast message in a LAN, how does spanning tree protocol solve this problem
6. given size of a frame is 100 bytes, what should be the datarate/bandwidth of the sender so that transmission delay equals propagation delay. Length of wire is 50km, propagation speed is 2x10^8m/s.

module-1 5m

1. What is a NAT device in ipv4. What problem does it solve and how?
2. differences between tcp and udp protocol

3. (a) explain the terms : subnet, network address, subnet mask
(b) list all the ip addresses in the subnet 192.168.20.73/3. explain with proper reason
(c) given that 192.168.20.73 lies in a subnet with subnet mask 255.255.255.240. will 192.168.20.81 lie in the same subnet?. Explain with reason

4. what is the size of the smallest possible subnet that can give seperate subnets for 4 divisions containing 60, 62, 12,120 devices respectively

5. Suppose a 128-Kbps point-to-point link is set up between Earth and a rover on Mars. The distance from Earth to Mars (when they are closest together) is approximately 55 Gm, and data travels over the link at the speed of light—3×10^8 m/sec.
(a)Calculate the minimum RTT for the link.
(b) Calculate the delay × bandwidth product for the link.
(c) A camera on the rover takes pictures of its surroundings and sends these to Earth. How quickly after a picture is taken can it reach Mission Control on Earth? Assume that each image is 5 MB in size.

6. Calculate the latency (from first bit sent to last bit received) for:
(a) A 1-Gbps Ethernet with a single store-and-forward switch in the path, and a packet size of 5,000 bits. Assume that each link introduces a propagation delay of 10 μs and that the switch begins retransmitting immediately after it has finished receiving the packet
(b) Same as (a) but with three switches.
(c) Same as (b) but assume the switch implements cut-through switching: it is able to begin retransmitting the packet after the first 128 bits have been received.

module-2 1m

1. What is a deterministic traffic model?
2. What is a stochastic process?
3. What is a probability distribution?
4. What is a probability density function (PDF) for a continuous random variable?
5. What is a probability mass function (PMF) for a discrete random variable?
6. What is Poisson distribution?
7. What is Pareto distribution?
8. What does a MAC address table contain?
9. What does a routing table contain?
10. What does a ARP table contain?

module-2 2.5m

1. why is a switch better than a HUB?
2. What is a Poisson distribution? What type of network traffic or event-arrival behavior can it model?
3. What is difference between switch and a bridge?
4. Given a continous probability distribution and its pdf, how to calculate its mean and variance?
5. Given a discrete probability distribution and its pdf, how to calculate its mean and variance?

module-3 5m

1. Explain everything about VLANs
2. create a simple network topology with a router with 2 ports connected to 2 different LANs, each LAN contains a single switch connected to 2 PCs, now for every necessary port give IP adddresses as needed, for every router give a routing table, and every switch a mac address table. Allocate IP addresses in such a way that the whole subnet we allot to this topology is as small as possible
3. in socket programming using C we done in lab, explain the flow for running a TCP communication between a client and a server, what all system calls we used, briefly explain
4. for file handling in C using open, read, write, close syscalls, explain all of these functions in detail
5. Explain about error detection , what layer is it used at? Explain CRC, FCS with examples briefly

module-3 1m
1. What is Quality of Service (QoS)?
2. What are QoS parameters?
3. What is a QoS architecture?
4. What is Integrated Services (IntServ)?
5. What is Differentiated Services (DiffServ)?
6. What is a scheduling algorithm?
7. What is FCFS scheduling algorithm
8. What is round robin scheduling algorithm?
9. What is priority based scheduling algorithm?
10. What is Weighted Fair Queuing (WFQ)?

module-3 2.5m
1. What is QoS? List the important QoS parameters and briefly state why they are important.
2. Explain token bucket method, where its used?
3. explain leaky bucket algorithm, where its used?
4. explain round robin scheduling algorithm in more detail
5. explain priority based scheduling in more detail

module-3 5m
1. explain the statement “traffic sometimes doesn’t come in uniform flow, it comes in bursts”. Explain how exactly token bucket lets this burstiness be processed if it comes only often not continously
explain for the example : bucket refill rate – 1token/s, bucket size 10tokens. For time t=0,1,2,….. the requessts come in this fashion : 1,0,0,4,2, assume each req needs 2 tokens and the max buffer size of requests is 5

you need to explain when/if bucket overflows and starts discarding requests, and every request what time it will be processed
2. explain exactly how leaky bucket ensures output flow is of constant rate, smooth
3. Consider five processes P1, P2, P3, P4, and P5 with arrival times 0, 1, 2, 3, and 4 units respectively, and burst times 5, 6, 3, 1, and 5 units respectively.
Using the Round Robin CPU scheduling algorithm with a time quantum of 3 units, draw the complete Gantt chart showing the execution order of all processes.

4. Consider five processes P1, P2, P3, P4, and P5 with arrival times 0, 5, 12, 2, and 9 units respectively, burst times 11, 28, 2, 10, and 16 units respectively, and priority values 2, 0, 3, 1, and 4 respectively.
Using the Preemptive Priority CPU Scheduling algorithm, where a lower numerical value indicates a higher priority, draw the complete Gantt chart showing the execution order of all processes.

In individual queues we can use FCFS
5. Consider five processes P1, P2, P3, P4, and P5 with arrival times 0, 2, 4, 6, and 8 units respectively, and burst times 5, 3, 8, 4, and 6 units respectively.
Using the First-Come, First-Served (FCFS) CPU Scheduling algorithm, draw the complete Gantt chart and calculate the average turnaround time.